

# MAP Estimation for a Univariate Gaussian with Known Variance

COM3031 Probabilistic Machine Learning

## 1 Problem Setup

Let  $\mathbf{y} = (y_1, y_2, \dots, y_n)$  be a set of independent and identically distributed observations drawn from a Gaussian distribution with unknown mean  $\mu$  and known variance  $\sigma^2$ :

$$y_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, \dots, n.$$

We place a Gaussian prior on the mean:

$$\mu \sim \mathcal{N}(\mu_0, \sigma_0^2).$$

Our goal is to compute the **maximum a posteriori (MAP)** estimate of  $\mu$ .

## 2 Likelihood

The likelihood function is

$$p(\mathbf{y} \mid \mu) = \prod_{i=1}^n \mathcal{N}(y_i \mid \mu, \sigma^2) = \left( \frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left( -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right).$$

## 3 Prior

The prior distribution over  $\mu$  is

$$p(\mu) = \mathcal{N}(\mu_0, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp \left( -\frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 \right).$$

## 4 Posterior and MAP Objective

By Bayes' rule,

$$p(\mu \mid \mathbf{y}) \propto p(\mathbf{y} \mid \mu) p(\mu).$$

Taking logarithms and ignoring constants:

$$\log p(\mu \mid \mathbf{y}) = \log p(\mathbf{y} \mid \mu) + \log p(\mu) + \text{const.}$$

The MAP estimate is defined as

$$\hat{\mu}_{\text{MAP}} = \arg \max_{\mu} [\log p(\mathbf{y} \mid \mu) + \log p(\mu)].$$

## 5 Derivative of the Log-Likelihood

$$\log p(\mathbf{y} \mid \mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 + \text{const.}$$

Taking the partial derivative with respect to  $\mu$ :

$$\frac{\partial}{\partial \mu} \log p(\mathbf{y} \mid \mu) = \frac{1}{\sigma^2} \left( \sum_{i=1}^n y_i - n\mu \right).$$

## 6 Derivative of the Log-Prior

$$\log p(\mu) = -\frac{1}{2\sigma_0^2} (\mu - \mu_0)^2 + \text{const.}$$

Taking the partial derivative with respect to  $\mu$ :

$$\frac{\partial}{\partial \mu} \log p(\mu) = \frac{\mu_0 - \mu}{\sigma_0^2}.$$

## 7 MAP Stationary Condition

Setting the derivative of the log-posterior to zero:

$$\frac{\partial}{\partial \mu} \log p(\mu \mid \mathbf{y}) = \frac{1}{\sigma^2} \left( \sum_{i=1}^n y_i - n\mu \right) + \frac{\mu_0 - \mu}{\sigma_0^2} = 0.$$

Rearranging terms:

$$\mu \left( \frac{n}{\sigma^2} + \frac{1}{\sigma_0^2} \right) = \frac{1}{\sigma^2} \sum_{i=1}^n y_i + \frac{\mu_0}{\sigma_0^2}.$$

## 8 MAP Estimate

Solving for  $\mu$ :

$$\hat{\mu}_{\text{MAP}} = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \left( \frac{1}{n} \sum_{i=1}^n y_i \right) + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0$$

## 9 Interpretation

- The MAP estimate is a weighted average of the sample mean and the prior mean.
- The weights depend on the relative uncertainty of the data ( $\sigma^2$ ) and the prior ( $\sigma_0^2$ ).
- **No data** ( $n = 0$ ):

$$\hat{\mu}_{\text{MAP}} = \mu_0.$$

- **Infinite data** ( $n \rightarrow \infty$ ):

$$\hat{\mu}_{\text{MAP}} \rightarrow \frac{1}{n} \sum_{i=1}^n y_i.$$